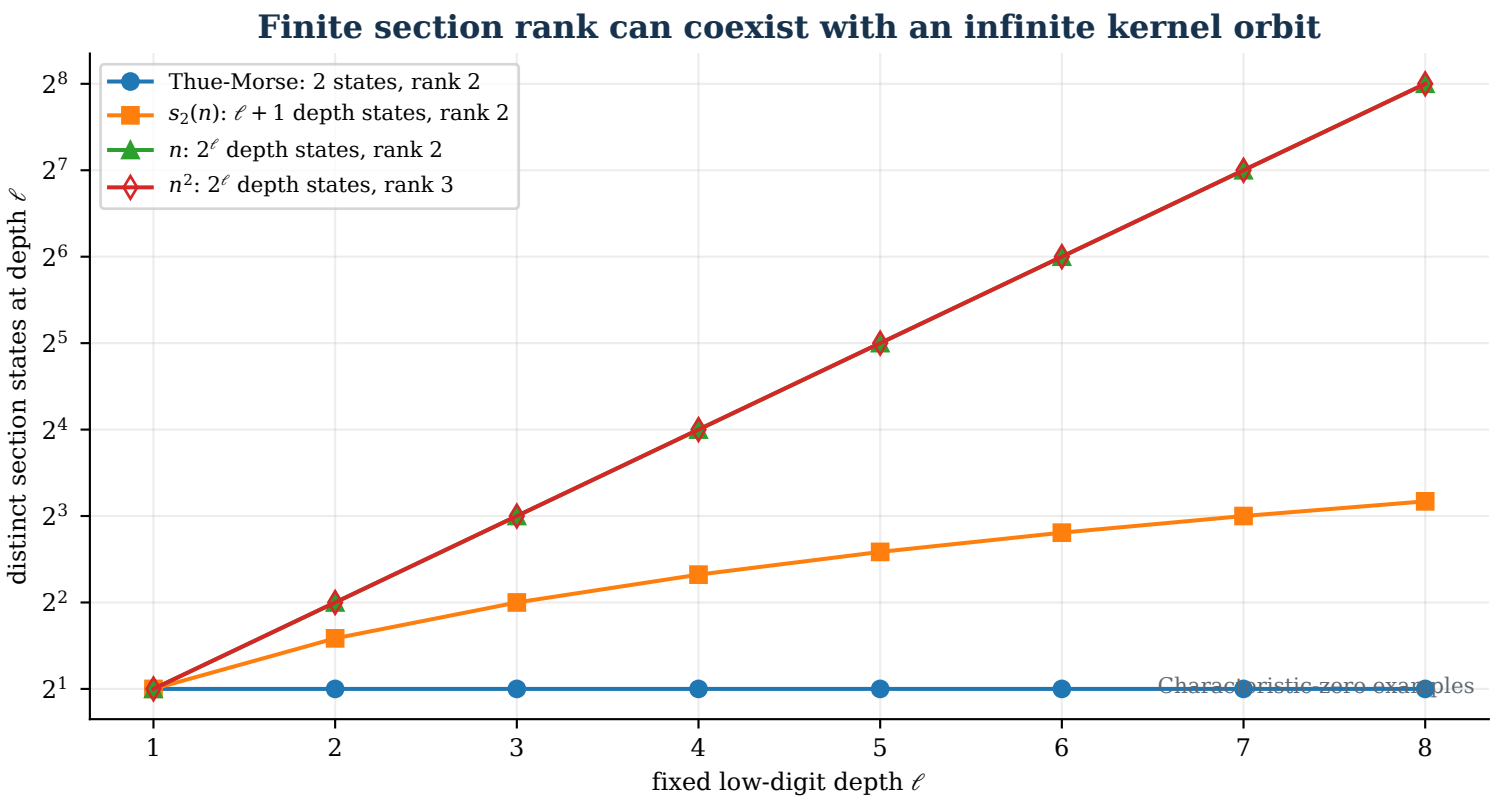
Finite section rank can coexist with an infinite kernel orbit
Thue-Morse: 2 states, rank 2
$s_2(n)$: $\ell + 1$ depth states, rank 2
n: 2^ℓ depth states, rank 2
n^2: 2^ℓ depth states, rank 3
distinct section states at depth ℓ
2^1
2^2
2^3
2^4
2^5
2^6
2^7
2^8
1
2
3
4
5
6
7
8
fixed low-digit depth ℓ
Characteristic-zero examples